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Training Load Monitoring in Soccer

Introduction

Examining the training process is essential to our understanding of why we measure training load. By measuring the training load we are looking to imply a dose-response relationship with the outcome parameter. The dose or the training load is the prescribed exercise and the response of interest in soccer being the associated fitness gain, fatigue accrued or injury risk. Examining the dose-response relationship in this fashion allows us to develop a knowledge of how a player may respond to a training dose and in future become more proactive and manipulate the dose rather than reactive and reacting to the response. It may help us produce a response we want (fitness gain) or prevent a response we would like to avoid (injury). The dose-response relationship is deemed a fundamental principle of training by the American College of Sports Medicine. It has also been suggested (E.W. Banister, 1991; Manzi, Iellamo, Impellizzeri, D'Ottavio, & Castagna, 2009) that a valid measure of “*training load*” should show a dose-response relationship with the training outcome. The training outcome is usually assessed periodically by measurement using an assortment of fitness tests, performance parameters or injury records. Why is the dose-response relationship so important to us as practitioners? Usually we react to a response, whether this is an injury or a fitness test score. Given that we want to avoid injuries and frequent fitness testing may be impractical, understanding the response to a given dose gives us the ability to be proactive in achieving our aims as coaches. Understanding the training process is essential to our understanding of what measurements of training load may show such dose-response relationships.

The Training Process

The training process has been conceptualized quite nicely by Impellizzeri and colleagues (Impellizzeri, Rampinini, & Marcora, 2005). Figure 1 below shows how both the prescribed training and the characteristics of the individual (genetics, training status etc) combine to form the internal training load.

###Insert Impellizzeri et al figure 2005 here###

Figure 1. The Training Process (Impellizzeri et al, 2005)

This is best explained by the example of two people of different fitness levels running a 10km race at the same pace, both finishing at the same time. The fitter person would find this less demanding internally and analysis of heart rate data would show it lower than the other person. In soccer related setting this also means that those players measured to have run the same distance will only show the same response if their personal characteristics are exactly the same. This scenario is highly unlikely, therefore as the model shows it is ultimately the internal training load that is the stimulus for training adaptation. Measuring training load is particularly difficult to achieve in sports such as soccer since different exercise designs will lead to different physiological and mechanical demands as well as inter-individual responses to the prescribed exercise. (Bangsbo, Mohr, & Krstrup, 2006)

The measurement of training load is often described as either internal or external. Internal training load monitoring is usually heart rate (HR) or rating of perceived exertion (RPE) based and is often calculated with the integration of time, intensity and a weighting factor. Intensity has been measured objectively using heart rate as it has been shown to have a linear relationship with oxygen consumption (Bot & Hollander, 2000), which is widely regarded as the gold standard measure of exercise intensity (ACSM, 2010). Consequently, the use of HR as a measure of intensity demonstrates validity. In recent years development of automated camera tracking systems, GPS and accelerometers have also brought the measurement of external training load to fore beyond the prescription of time and a number of actions from the

coach. The aim of all these method is the same; this is to describe the training dose as a single function of variables such as frequency, intensity and time that have long been used to manipulate the training dose.

In this chapter the methods used to examine both internal and external loads will be explored and the validity of these methods assessed. Furthermore in a practical or applied environment certain factors may influence what you can and cannot apply. These could include costs, time and practicality. This will also be summarised for each method. The decision as to whether a method is worth the cost and time is an individual one to make which may be determined by resources available but should also be governed by the validity of each method. It is my hope that this chapter helps coaches and practitioners decipher all these details to help them in this area.

Internal Load

Banisters TRIMP

Banister et al (1975) were one of the first in the pursuit of a solitary number to describe the training load or training impulse (TRIMP). He had originally proposed a three zone model where exercise was categorized as low, moderate or high intensity and each zone was weighted 1, 2 and 3 accordingly. However the method Banister (1991) developed as TRIMP that is widely used today is based on heart rate and a modelled blood lactate response to increasing intensity of exercise. Banister's TRIMP takes into consideration the intensity of exercise as calculated by the heart rate (HR) reserve method and the duration of exercise. The mean HR for the training session is weighted according to the relationship between HR and blood lactate as observed during incremental exercise and then multiplied by the session duration.

TRIMP is calculated using the formula below:

$$\text{time (mins)} \times \Delta\text{HR} \times y$$

Where;

t = duration (mins)

ΔHR = fractional elevation in HR or HR reserve

y = weighting factor

The ΔHR is weighted in such a manner that it reflects the intensity of effort as a guard against giving a disproportionate importance to long durations of low intensity exercise compared with more intense exercise. The multiplying factor (y) weights the ΔHR according to the classically described increase in blood lactate in trained male and female subjects, respectively.

Banister used the TRIMP to model endurance performance by using the TRIMP as measure of training load from which he modelled the dose-response relationships with fitness and fatigue. Banister theorized that each training bout produced both a fatigue and a fitness impulse and that fatigue decays three times faster than fitness, hence training adaptation and enhanced performance. Performance at any given time is a result of the fitness level less the accrued fatigue. Morton et al (1990) modelled endurance performance for two athletes using Banisters TRIMP. These results gave Banisters TRIMP credence in endurance events.

The modelling conducted to date has focused on endurance athletes with long training schedules who need to optimize performance for a relatively short competition period from 1 day (e.g. marathon) to a few weeks (e.g. cycling tour). The modelling of performance in endurance sports (Morton, 1990) somewhat validates Banisters TRIMP. However the modelling process has been subject to modifications (Busso, 2003) for improvements in predictions.

There are two major limitations in using Banisters TRIMP in intermittent sports such as soccer. Firstly the use of mean HR may not reflect the fluctuations in HR that occur during intermittent exercise. The mean exercise intensity in soccer matches has been widely reported to be around the anaerobic threshold at 85% of HRmax (Stolen, et al., 2005) but has also been reported to peak at intensities close to HRmax (Ascensao et al., 2008). Secondly, the use of generic equations for males and females implies that the gender is the only factor making athletes different and doesn't necessarily take into consideration individual differences that effect training load that the Impellizzeri et al (2005) model implies.

Associated Costs: HR monitors

Practicalities: Easy to calculate once correct data has been identified and downloaded, same formula for each player. Some software provides this calculation. Availability of match data a problem at senior levels as HR monitors may not be worn. However can be applied at all other levels.

Edwards TRIMP

Edwards (1993) proposed a zone based method for the calculation of training load. The time spent in five pre-defined arbitrary zones is multiplied by arbitrary coefficients to quantify training load. The proposed the zones based on HRmax with 10% zone widths and corresponding coefficients as can be seen in the table 1 below.

HR Zone (% HRmax)	Weighting Factor
50-60%	1
60-70%	2
70-80%	3
80-90%	4
90-100%	5

Table 1. HR weightings proposed by Edwards (1993)

This method gained popularity as the default setting on a popular HR monitor system. However the coefficients are void of physiological underpinning and the zone limits remain predefined and void of any metabolic or physiological performance thresholds. Such zones and weightings would imply the training adaptation in zone 5 is five times greater than in zone one and that the relationship between training intensity. However no study to date has proven this to be case. The weightings used by Edwards (1993) are not validated through a relationship with a known physiological response. Neither has a training study looking at the quantification of the training from this method been conducted to assess the dose-response relationship. Through this chapter we will come across a number of methods that have been considered valid through its relationship with this method. This has been done on the basis that heart rate is valid measure of intensity (Bot & Hollander, 2000), but where intensity is only one term in the equation for training load (the others being “time” and a “weighting factor”), how can the validity of training load be assumed just because heart rate is a valid indicator of intensity only.

On the other hand there is evidence to support the use generic high intensity thresholds. Castagna et al (2011) showed a dose-response relationship between the time spent above 90% and changes in fitness. Although useful such approaches used in isolation risk ignoring the training load accrued from training below such thresholds and the remaining intensity continuum. This risks vital load information being missed which could be the difference between fitness and injury.

Costs: HR monitors

Practicalities: Usually weightings and zones can be set in software that comes with hardware therefore calculation is relatively easy. Availability of match data a problem at senior levels.

Lucia's TRIMP

Lucia, Hoyos, Santalla, Earnest, & Chicharro (2003) based their measure of training load around the 1st and 2nd ventilatory thresholds (VT1 & VT2). The method provides three zones: low (<VT1), moderate (VT1 - VT2) and high (>VT2). Each zone is given a coefficient of 1, 2 and 3, respectively. Time spent in each zone is

multiplied by the relevant coefficient and summated to provide a TRIMP score. However, like Edwards (1993), the weightings are not based on any scientific evidence and/or physiological data. Earlier work by Banister et al (1975) with swimmers used the same weighting coefficients (1, 2, and 3) for low, moderate and high-intensity work, however he changed his approach to later base weighting on the blood lactate response. This sort of weighting implies that exercise at high-intensity is three times as demanding as exercise at low intensity. Lucia used this method to successfully compare the training load distribution in two different cycling tours. They reported no significant difference in the training load calculated by their method for two different cycling tours (Vuelta a Espana compared to the Tour de France). Training using this three-zone model in endurance sports has received some attention (Esteve-Lanao, Foster, Seiler, & Lucia, 2007; Seiler & Tonnessen, 2009), giving the method credence due its use in elite settings. Seiler described the polarized training methods popular with endurance athletes where ~80% of their training time is spent in zone 1 (<VT1). The metabolic thresholds used to identify the zones have shown to relate well to endurance performance (Amann et al., 2006). However the weightings remain arbitrary. This system appears to be best used by monitoring the time in each zone and the distribution in competition and training. This by no means dictates that a universal score from the associated coefficients is valid. Furthermore the weighting of each zone implies, that the training adaptation would be the same regardless of where in the zone an athlete trained. For example if the threshold for VT2 is identified at 85% of HRmax a training session with an intensity of 95% of HRmax would be given the same weighting as a training session at 85% of HRmax but different to if the athlete trained at 84%. The study of Denadai et al (2006) shows how a 5% difference in training intensity (95%vVO2max vs. 100%vVO2max) produced different training adaptations. For these reasons I would question the use of any zoning method. Impellizzeri et al (2005) did demonstrate that training below and above such thresholds may produce differing training responses in soccer players (although the threshold was based on an arbitrary lactate value). But monitoring just high intensity activity in isolation means that the accrued training load from below the high intensity threshold could end up being ignored. To date no training study on Lucia's method has been conducted to validate it by demonstrating dose-response relationships in soccer.

Costs: HR monitors, regular testing and analysis.

Practicalities: Usually weightings and zones can be set in software that comes with hardware therefore calculation is relatively easy. Testing and interpretation of data for a whole squad is time consuming and can be costly. It would either require purchase of gas analysis systems/lactate analyzers or paying other parties to do the testing. Availability of match data a problem at senior levels.

Stagno's $TRIMP_{mod}$

Stagno, Thatcher, and Van Someren (2007) developed a modified version of Banister's TRIMP in an attempt to quantify training load for field hockey. Rather than use a generic equation to reflect a hypothetical blood lactate profile, these authors directly measured the blood lactate profile of the hockey players. The weightings they used therefore reflected the profile of a typical blood lactate response curve to increasing exercise intensity for the specific population, in this case the hockey team. While not truly individualized, their method used the mean blood lactate profile from all of the players to generate the weightings, providing at least some degree of individualisation. They then anchored five HR zones around the lactate threshold (LT, 1.5 mmol) and the onset of blood lactate accumulation (OBLA, 4 mmol), with the resulting zone weightings being 1.25, 1.71, 2.54, 3.61 and 5.16. The accumulated time in each HR zone was then multiplied by its respective zone weighting to derive an overall $TRIMP_{mod}$. The research quantified the TL in hockey and established relationships between $TRIMP_{mod}$ and various fitness parameters during the course of a season.

They found the mean weekly $TRIMP_{mod}$ shared significant relationships with changes in running velocity at vOBLA and VO_{2max} . They also reported significant correlations between time spent in high-intensity activity and the change in VO_{2max} and the change in vOBLA. The dose-response relationships of this study suggest that $TRIMP_{mod}$ is a method by which TL could be measured. The original TRIMP (Banister, 1991) is calculated using the mean HR for a particular exercise session or interval of training. Stagno, used time accumulated in zones to reflect the different

intensities team sport players work at in comparison to endurance athletes may work at a similar intensity for longer periods. This may make the use of a mean HR more suitable and less of an issue in endurance athletes. But modern day training regimens with endurance athletes also use high intensity interval type approaches. Stagno didn't compare his method to Banisters and we do not know if there is any significant difference when using the zone method of Stagno compared to the mean HR method as used by Banister. It has been shown that the difference is significant from a single exercise bout (Akubat, 2009). The zones used by Stagno were based on the HR at lactate threshold (LT), defined as 1.5 mmol.L⁻¹ and the onset of blood lactate accumulation (OBLA) defined as 4 mmol.L⁻¹. They used the blood lactate responses at four different speeds from their player sample to create an equation for the weightings or the 'Y' value as defined by Banister. Zones 2 and 4 were created around the mean HR at LT and OBLA. A zone width of 7% fractional elevation was formed at these points. Zones 1, 3 and 5 were then created around zones 2 and 4. The pre-requisite for the use of this method was that the HR at LT and OBLA for all players fell within the created zones. I have found with a larger squad of players this pre-requisite could not always be met. The use of zones also still holds the limitation of giving the same weighting to exercise spanning the whole zone. For example if a zone which was 70-80% of HRmax had the same weighting, an athlete training at 71% would get the same weighting as someone training at 79%. It is difficult to ascertain however if this difference would affect physiological adaptation, and there appears to be no study to date that has examined this fundamental training question. However, with the method of Stagno, the zones are created around the thresholds, so there may well be a situation where players exercising in the same zone, gaining the same weighting, are working above and below a metabolic threshold. Impellizzeri et al (2005) showed this could produce different results. It must be highlighted that although the zones are based on metabolic criteria, they are created with arbitrary values of lactate and are therefore not individualised as Lucia's are. Another issue with using weighting individualised to a team is that it still does not account for individual differences. Figure 2 below shows data collected from a squad of professional players from the championship division in England. Note how the regression line from which the weightings are calculated is well below the data points for some players especially at higher intensities, possibly leading to an

underestimation of training load.

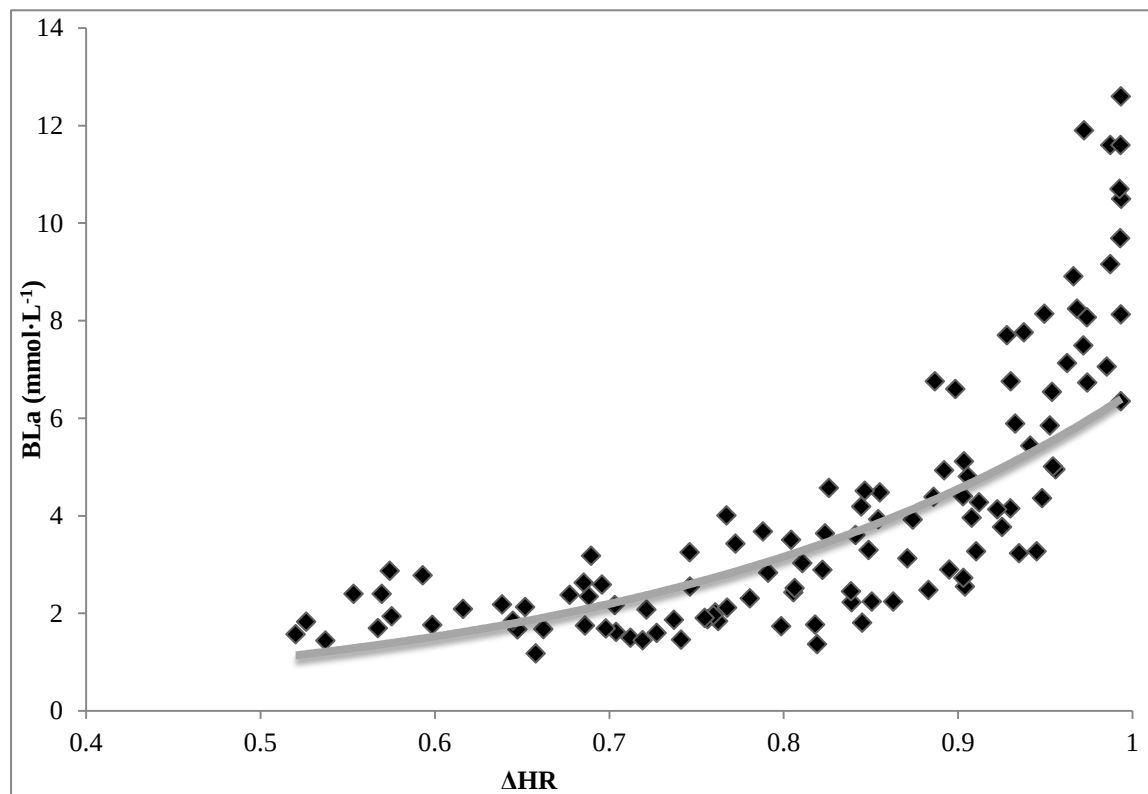


Figure 2. Blood lactate responses for a squad of championship footballers.

In summary the work of Stagno highlighted some of the complexities involved in monitoring training load in team sports. They highlighted the need for specific weightings, although they did not fully individualise these. Their use of zones was an attempt to move beyond the use of mean HR. The limitations highlighted exist, however they produced significant dose-response relationships and at the time this study was a big step in the right direction. A number of teams have used the Stagno weightings, but often the point is missed. These weightings are specific to that team and to use this approach successfully you would have to do the same testing to produce the weightings specific to your team.

Banister used mean heart citing calculating the TRIMP for each reading would be too problematic. However modern computing does not allow for that being an excuse anymore. Therefore given the limitations of zones as highlighted above why not assign each HR reading a weighting, thereby circumventing the need for zones. The next method does exactly that.

Costs: HR monitors, regular testing and analysis.

Practicalities: Usually weightings and zones can be set in software that comes with hardware therefore calculation is relatively easy. Testing and interpretation of data for a whole squad is time consuming and can be costly. It would either require purchase of lactate analyzers or paying 3rd parties to do the testing. Availability of match data a problem at senior levels.

Individualised TRIMP (iTRIMP)

A group of Italian researchers mainly led by *Vincenzo Manzi* are the first and most prominent researchers of this method. They have conducted the majority of research with this method which I believe shows greater validity than other methods of internal training load and deserve great credit for advancing the work in this area from which we all now benefit.

Using this method the iTRIMP weighting is based on the individual's own heart rate–blood lactate response to incremental exercise, as measured during a standard lactate threshold test protocol. Furthermore, as a development on previous methods *Manzi, et al., (2009)* did not use HR zones or mean heart rate. The iTRIMP scores were calculated for each HR reading and summated to give an overall iTRIMP score. In comparison to methods employing zones this meant they had effectively created a zone for each HR reading from *HRrest* to *HRmax*. Therefore, these authors individualised the weighting to the athlete, which goes beyond the individualisation by gender (*Banister, 1991*) or group (*Stagno et al., 2007*). Moreover, the iTRIMP weighting is not arbitrary as in the case of *Edwards (1993)* and *Lucia et al (2003)*. Consequently, this method overcomes many of the limitations of previous methods.

Manzi et al (2009) who published results using iTRIMP method showed that after an 8 week period of training in recreational runners, the mean weekly iTRIMP significantly correlated with changes in velocity at lactate threshold (vLT ; $r=0.87$) and velocity at the onset of blood lactate accumulation ($vOBLA$; $r=0.72$). The mean weekly iTRIMP also showed significant correlations with changes in 5000m ($r=-0.77$) and 10000m ($r=-0.82$) running performance. However, *Manzi* reported that *Banister's*

TRIMP didn't show significant relationships with any fitness parameters or performance measures. The participants used in their study were recreational runners and therefore the effectiveness of the iTRIMP method needed to be also measured in team sports where the activity is highly intermittent. As the HR-VO2 relationship appears to be valid even during intermittent exercise (Esposito, Impellizzeri, Margonato, Vanni, Pizzini, & Veicsteinas, 2004). Therefore iTRIMP method has potential for use within a soccer environment.

The iTRIMP method at this point was shown to be the only method that based its weightings on the individual response and which is based on a physiological parameter, a combination which is evidently absent in the previous methods discussed.

Recently two studies were published looking at the iTRIMP in senior and youth soccer players, respectively (Akubat, Patel, Barrett, & Abt, 2012; Manzi, Antonio, Maria, Ivan, & Carlo, 2012). It was found in pre-season with an Italian Serie A team, iTRIMP showed dose-response relationships with changes in VO_{2max} ($r=0.77$), velocity at ventilatory threshold ($r=0.78$), $vOBLA$ ($r=0.64$) and Yo-Yo IR1 performance ($r=0.69$). The study conducted in-season with a youth team from an English championship club showed a dose-response relationship with changes in vLT ($r=0.67$) but not with $vOBLA$. Akubat et al (2011) also compared the weekly iTRIMP scores to session RPE and Banisters TRIMP and a modified version of Stagno's TRIMP (group weightings but no zones). The results were similar to previous studies comparing methods to each other. Session RPE related to Banisters TRIMP, however iTRIMP showed poor relationships with the other measures. This showed that the method that does show dose-response relationships does not relate that well to those methods employed previously.

This is all great new information, but the versatility of this method is best shown in the study by the same group of researchers this time headed by Iellamo et al (2012). They employed the iTRIMP method to measure training load with cardiac patients. They compared the effects of a continuous aerobic training and interval aerobic training programmes on numerous health and fitness measures. They matched the dose of exercise for each group using the iTRIMP method. There was no significant difference between the groups in all the measured variables. The

measurements did show changes over the training period, for example VO_{2peak} improved by 22%. But there was no difference between the groups. Therefore so long as the iTRIMP was the same the distribution of the training on the intensity continuum appears irrelevant. This has major implications for practices that could be employed in different sporting situations. In summary the iTRIMP research since 2009 has progressed the work in training load monitoring forward considerably.

Costs: HR monitors, regular testing and analysis.

Practicalities: Software is now available that allows quick and easy analysis of HR and provides iTRIMP scores, however manual calculation is time consuming. Testing and interpretation of data for a whole squad is time consuming. It would either require purchase of lactate analyzers or paying 3rd parties to do the testing. Availability of match data a problem at senior levels.

Session RPE

Training load as measured by the session RPE is a subjective method of quantifying the load placed on an athlete. It is calculated by multiplying the session intensity by the duration to provide a measure of load in arbitrary units. The intensity is described as a number (0-10) on the CR-10 Rating of Perceived Exertion (RPE) scale proposed by Borg, Hassmen & Lagerstrom (1987). The significant relationships between RPE and other measures of intensity like HR ($r=0.89$) and plasma lactate concentration ($r=0.86$) have been demonstrated (Gabbett and Domrow, 2007). However just because it is a valid measure of “intensity” doesn’t imply that it will be a valid measure of “load”. Foster et al (1996) showed that increasing the TL (measured by Session RPE) tenfold over 12 weeks resulted in a 10% improvement in performance with runners and cyclists. However the study also showed poor dose-response relationships between session RPE and changes in performance ($r=0.29$). Gabbett and Domrow (2007) also reported a poor association between session RPE and changes in skinfold thickness, speed or maximal aerobic power during any of the training phases they monitored. In many sports the usefulness of session RPE emerges from its ease of use compared to the technical nature of using HR monitors. Issues include loss of data and player compliance. The study of Impellizzeri

et al (2004) compares session RPE to the methods of Banister (1991), Edwards (1993) and Lucia et al (2003) and the relationships imply the validity of session RPE. However the limitations of these methods detailed earlier in this chapter would make the far from criterion measures. Impellizzeri et al (2004) conclude that with only 50% of the variation in session RPE being explained by heart rate it cannot be deemed a valid substitute. A similar study was conducted by Alexiou and Coutts (2008) where session RPE was correlated in different types of training sessions for women soccer players. Both studies reported significant relationships with HR-based methods. Such approaches where new methods are compared to older methods to validate their use are becoming quite common. But relationships presented in such studies could imply that the method is just as poor not only just as good.

More recently Brink et al (2010) assessed the 'dose-response' relationship of session RPE with performance and recovery. They used session RPE, total quality of recovery (TQR) and performance in young elite soccer players over a whole season. Daily logs were kept by players and coaches to report TL after sessions and TQR prior to the next session. To assess performance they used the interval shuttle run test (ISRT) on a monthly basis. They applied multi-level modelling techniques to examine if session RPE could predict performance and recovery outcomes. They reported that the number of training days significantly predicted the performance outcome as represented by a decrease in the HR during the ISRT. However the model did not significantly predict performance with session RPE or TQR. Although the simplicity of session RPE cannot be denied, the usefulness of the information it provides may be questioned. The study of Brink et al (2010), Gabbett and Domrow (2007) and Foster et al (1996) show that session RPE doesn't appear to fit dose-response models.

Although dose-response relationships with fitness or performance appear to be lacking a recent study with looking at the changes in session RPE within the individual has shown that it maybe useful as a predictor of injury. Rogalski et al (2013) found that larger weekly or fortnightly session RPE scores and larger week to week increases in load significantly related to injury risk. Given that injuries cause untold problems to the individual, the coach, the team and financially and the inexpensive nature of session RPE if it helps in the prevention of injury its use for this reason alone is vindicated. Remembering that session RPE is a subjective

measure of training load, examination of why this may be seen should also be considered. Why does session RPE help relate with injuries and not performance or fitness responses? One of the main precursors for injury is fatigue. Unpublished data from a study I did may help explain this relationship. Table 2 below shows both internal and external training load measures from two simulated soccer matches (Akubat et al, unpublished) using the modified BEAST protocol (Akubat et al, 2013).

	iTRIMP		Distance		HID		Player Load		sRPE	
Match	1	2	1	2	1	2	1	2	1	2
Mean	409	304	10810	10604	3336	2868	1301	1257	639	728
SD	174	91	664	592	718	754	94	145	99	99
P value	0.015		0.171		0.046		0.426		0.010	
Cohen's D	0.48		0.21		0.43		0.27		0.60	
Effect Size	Small		Small		Small		Small		Moderate	

Table 2. Load measures from two consecutive matches 48 hours apart.

These matches were split by 48 hours and the data on high intensity distance covered shows the external performance was significantly lower. Players were instructed to run as far as they could therefore it could be reasonably assumed that fatigue or a lack of recovery could be the reason for their reduced performance. However the interesting change is in the iTRIMP and sRPE scores. iTRIMP would suggest a reduced internal load whereas session RPE would suggest an increased perception of internal training load. In this situation we are dealing with objective versus subjective data. It is my belief that the higher sRPE score is a reflection of the perception of load in an unrecovered state or under fatigue. Therefore in situations where there is inadequate recovery the same or reduced external load will be perceived as greater internal with sRPE. So in the situations described by Rogalski et al (2013) where large week to week increases in sRPE predict injury could this increase in sRPE be due to lack of recovery and the increased injury risk due to levels of fatigue? Elloumi et al (2012) have demonstrated relationships between sRPE and fatigue measures. Could such subjective measure be more reflective of fatigue than training load? It is unlikely that such spikes in week to week training load are programmed by coaches. The answers to these questions require more research and those interested in the area could maybe pursue such research questions.

A recent modelling study with runners (Wallace, Slattery, & Coutts, 2013) has demonstrated that in runners session RPE may prove valuable in the prediction of performance along with Banisters TRIMP. Indeed Banisters TRIMP has long been used in modelling and predicting endurance performance (E.W. Banister, Carter, & Zarkadas, 1999; Busso, 2003; Busso, Carasso, & Lacour, 1991; Busso & Thomas, 2006; Morton, 1990). Translation of such studies into intermittent sports is the next step for validating and deciphering the superior methods for training load monitoring in intermittent sports.

Costs: minimal

Practicalities: Easy to administer and simple calculation. Can be difficult to stop one player influencing scores of another player in a team environment and can add to poor reliability and variability that has been reported. Software and smartphone apps now available to make this easier. Match data available.

External Load

The measurement of external load in soccer goes as far back as 1952 (Winterbottom, 1952) when hand notation methods were used to estimate the external demands of a game through the use of distance. In more recent history the use of automated camera systems brought the use of distance and break downs of the velocities at which distances were covered to the fore. This method allowed determination of positional differences (Di Salvo, Baron, Tschan, Montero, Bachl, & Pigozzi, 2007), levels of play (Mohr, Krstrup, & Bangsbo, 2003) and also the match to match variation (Gregson, Drust, Atkinson, & Salvo, 2010). More recently much of this research has been again examined using a newer method. The invent and use of GPS technology has revolutionised the way we track, monitor and examine both the loads on players and performance. Whereas semi automated camera systems where limited to stadia in many instances GPS technology has the advantage of being able to be used almost anywhere and providing real-time data too. Ironically in some stadia due to surrounding structures GPS units may struggle to connect to the satellites they rely on to measure their location. The use of these devices has also brought many challenges as they are also now equipped with accelerometers, gyroscopes, magnetometers meaning the wealth of data that you can end up with is considerable.

Instantaneous, high frequency measurements has led to many questioning the use of HR based measures. One of the most common criticisms is that HR doesn't react to short high intensity movements where GPS and accelerometers may do. This is a fair criticism but one that is often overstated because you will see HR rise and remain elevated between bouts in repeated high intensity efforts. When the accelerometer or GPS measures zero when the players is stood still the HR is still elevated minimising any such underestimation. Also as the HR calculations use time, and time in match-play accounts for 2% for sprinting activities (Bangsbo, 1994). Therefore although the distance covered at maximal and supramaximal intensity makes up a considerable percentage of the distance covered the "time" at these intensities which is what is used in calculations being only 2% minimises any underestimation.

The advent of such technology has also led to an emergence of data mining where everything possible is collected and data then examined by powerful analytical tools. On the other hand some practitioners favour the approach of using only the data that gives them useful information. Again the approach you take will depend on the resources at your disposal and the effectiveness with which you can use the data. Does it impact on fitness, performance, well-being and/or injury risk. In this section we will examine what external load can tell us about the dose-response relationship and assess the validity of external methods for the measurement of training load.

Distance

As an external measure of training load any dose-response relationship will be mediated by the individuals characteristics in accordance with figure 1 presented earlier in this chapter. A theoretical comparison of an elite runner and a recreational runner who run 5000m at the same pace exemplifies this. The stress on each runner will be determined by their individual fitness status and therefore even if the distance and the pace is the same the dose and therefore the response will be different in each individual. As fitness can also change within the individual that same 5000m distance at the same pace will not always produce the same level of stress.

Typically in soccer we have seen the use of total distance and high intensity distance (HID) as measure of external load or performance. High intensity distance has gained some credibility as a measure of exertion and performance through construct validity when comparing moderate level to elite level players (Mohr, *et al.*, 2003) (It has also been shown to vary greatly from game to game (Gregson, *et al.*, 2010). It has been suggested TD and HID maybe valid indicators of load as soccer players will run as far as possible in games. Others argue it is stimulus driven expenditure determined by factors such as state of play, position and tactics to name a few (Impellizzeri, *et al.*, 2005; Rampinini, Coutts, Castagna, Sassi, & Impellizzeri, 2007; Rampinini, Impellizzeri, Castagna, Coutts, & Wisloff, 2007). In assessing load using distance data speed thresholds have also been individualized.

Individualisation of Speed Thresholds

As each individual may possess different levels of fitness and physical capabilities comparing variations of distance data between players may give you comparable statistics on performance. However the factor of interest in this chapter is exercise dose and generic thresholds for speed will not give you comparable exercise dose data. The methods described below are those that have been popularized recently with training regimens in soccer.

Abt & Lovell (2009) proposed that high intensity speed thresholds should be individualised based on ventilatory thresholds in a similar manner to the methods proposed by Lucia et al (2003) discussed earlier. This would require a laboratory treadmill test with gas analysis. Abt & Lovell (2009) found that the median high intensity threshold as determined by the velocity at the 2nd VT or RCP was 15 kmh (range 14-16 kmh), which is considerably lower than many GPS and camera tracking system defaults. They found the mean distance run at high-intensity based on the default (19.8 kmh) and VT2 speed was 845 m and 2258 m signifying almost a 3 fold increase in the high intensity exercise dose. A follow up study (Lovell & Abt, 2013) found that total distance covered in the three different zone proposed by Lucia was 26%, 57%, and 17% for low, moderate and high intensity zones respectively. They also identified a 41% difference in the high-intensity distance covered between 2 players of the same positional role when zones were individualized compared to when zones were not (5-7%).

Given that speeds above high intensity thresholds around 15kmh encompasses a large zone as peak speeds have been reported to be well above 25 kmh (Gregson et al, 2005) other researchers have proposed the use of maximal aerobic speed (MAS) and peak speed to generate training zones. Using this method supra- maximal running (speeds >MAS) speed can be identified (Buchheit, Simpson, & Mendez-Villanueva, 2013) and used for programming, breaking down the high intensity running zone into smaller blocks than just above VT₂. When MAS has been used for individualized training regimens it exercise intensities have been prescribed as percentages of MAS (Baker, 2011)

Using such approaches where certain zones are individualized in such a way based on physiological thresholds the intensity becomes relative to the individuals' fitness

levels. Given the framework proposed by Impellizzeri et al (2005) this then could be considered an internalised measure of training load with the measurement of external performance. It would appear that if you are using external measurements for monitoring load that such individualisations are essential. What is difficult with such approaches is to assign a solitary number to describe the accrued training load. Numerous studies have shown the benefit of high intensity training programmes in soccer (Hoff, Wisloff, Engen, Kemi, & Helgerud, 2002). Castagna et al (2011) showed how the internal training load at high intensities shows dose-response relationships however such relationships with external performance have yet to be shown. There is likely to be a similar relationship given that training above VT2 or MAS would produce high heart rates. It is also worth giving consideration to the study of Denedai et al (2006) who showed marked contrasts in adaptation between groups that trained at 95% & 100% of $v\text{VO}_{2\text{max}}$. Hence the debate around zones and differences remains where, in a zone with a 10% width (e.g. 90-100%) a player exercising at 91% would get the same credit as one exercising at 99%. These approaches somewhat internalise external measures of load, but it still remains difficult to equate the distances covered in all zones irrespective of how they are defined into a solitary number for exercise dose or training load.

There are two major criticisms of using velocity based measures of load that have been used by GPS companies and researchers to promote other measures.

- 1) Movements that don't create vertical displacement are not accounted for.
- 2) Activity isn't considered high intensity unless speeds thresholds are breached where as accelerations without reaching top speed are just as if not more energetically demanding (Gaudino, Iaia, Alberti, Hawkins, Strudwick, & Gregson, 2013; Gaudino, Iaia, Alberti, Strudwick, Atkinson, & Gregson, 2013; Osgnach, Poser, Bernardini, Rinaldo, & di Prampero, 2010)

This has led to the development of both accelerometry derived load and metabolic power calculations both now available with some GPS technology suppliers.

Accelerometry

Accelerometry in soccer has come in conjunction with GPS technology as these units have had accelerometers incorporated into them. Accelerometer derived load has been described as external load, mechanical load and as centre of mass acceleration in all three planes of movement. The algorithms used are different depending on the GPS company. In the research as a load measure studies have sought demonstrate validity of accelerometry derived load by correlating this to some of the measures of internal load mentioned earlier. The Player Load (Catapult) and Body Load (GPSports) are two of the accelerometry derived load measures. Player Load appears to relate to sRPE & Edwards TRIMP (Casamichana et al, 2013) and Body Load doesn't appear to show a relationship with sRPE (Gomez Piri et al, 2012). Unpublished data (Akubat et al, unpublished) would suggest the Player Load shows very large correlations with sRPE ($r=0.77$) but only small relationships with iTRIMP ($r=0.16$). You could speculate that given iTRIMP has shown dose-response relationships (Akubat, *et al.*, 2012; Manzi, *et al.*, 2012; Manzi, *et al.*, 2009) the lack of a relationship with iTRIMP would bring the ability of Player Load to show a dose-response relationship into question. However this is speculation and to truly assess the validity of any accelerometry derived load measure a training studies such as those done previously are required (Akubat, *et al.*, 2012; Manzi, *et al.*, 2012; Wallace, Slattery, & Coutts, 2013).

Metabolic Power

Accelerations and decelerations even at low absolute velocities are high intensity and energy demanding activities. However the way high intensity activity is determined using velocity thresholds may not account for this (Gaudino, *et al.*, 2013; Gaudino, *et al.*, 2013)

Di Prempere et al (2005) developed a mathematical approach to quantify the estimated energy cost associated with any instant change in velocity. It was proposed, accelerated running on a flat terrain is considered energetically equivalent to uphill running at constant speed (Minetti, Moia, Roi, Susta, & Ferretti, 2002). Metabolic power is calculated as the instant energy cost multiplied by the instant

velocity. The ability to calculate instant “energy cost” based on known and measured data and the measurement of “instant velocity” from GPS allows this estimate of metabolic power to be calculated. As both velocity and acceleration are used it is argued this provides a better estimation of high intensity activity (Gaudino, *et al.*, 2013). Gaudinho et al (2013) showed that when the equivalent high intensity metabolic power threshold (a metabolic power of 20w.kg is considered the equivalent of running at a constant speed of 14.4kmh) is used in analysis of soccer training the actual high intensity activity could be underestimated by as much as 84 ± 54 %. Therefore the availability of metabolic power calculations through GPS software has been an interesting addition to the monitoring of load. As this chapter is being written no published studies have done a training study to assess dose-response relationships in soccer nor has metabolic power been compared to other methods. There are some theoretical limitations that must be considered. Acceleration measurement with GPS units at higher velocities has been shown to display increasing error (Akenhead, French, Thompson, & Hayes, 2013) and there has also been wide inter-unit variability reported (Buchheit, Al Haddad, Simpson, Palazzi, Bourdon, Di Salvo, & Mendez-Villanueva, 2013). There are also inter-individual variations in energy cost as the same acceleration velocities may not produce the same energy cost when data is examined closely (Gaudino et al, 2013).

Hopefully over the next few years research in this area will clarify this methods usefulness in the training process. But the underestimation of high intensity activity in the proportions demonstrated has serious implications from a training, adaptation periodization, fatigue and ultimately injury prevention when compared to the more traditional interpretation of high intensity activity based on velocity thresholds alone.

Match Load as a Measure of Training Load

The use of GPS and its associated technology has allowed the measurement of match-play activities in depths previously unknown. One of the approaches often used in practice but not really in research is that of using percentages of match based metrics to periodize and optimise training. For example, a player may aim to do two games worth of work (distance, accelerometry derived load or metabolic power) in a given week. This approach has fundamental flaws. Firstly to imply this

method works we must have evidence that a certain amount of match like activity produces a certain response. Secondly the variation in match-loads between games and positions (Gregson et al, 2010) would mean it is difficult to make assumptions about the typical match for a given player and competitive match play currently doesn't allow players to wear GPS devices. The question that arises with such an approach is are we trying to make the technology useful using match demands just because it is something we can now measure or does this method provide a measure of training load. Such a method determines load irrespective of information on initial and changing fitness levels. Therefore with respect to the Impellizzeri model there is little appreciation of the individual in the use of such a method. Rather the assumption is made that a certain match profile in a certain position is equally stressful irrespective of fitness status.

Costs: GPS units very expensive for whole squad

Practicalities: Player compliance issues for wearing units in some cases. Lots of data, which is actually useful? Once data is collected analysis/analysts required. Availability of match data a problem at senior levels.

Maximising Performance using Training Load Monitoring

The purpose of any player monitoring is to gain information to help understanding of the response and to produce this response when required. Soccer players may face different challenges to those in other sports where competition is infrequent which allow training a taper periods. In soccer the ideal scenario for a coach is to have a player able to participate to his maximal capability on a number of occasions. In many European leagues successful team regularly play 2-3 times in a week. To help maximize or understand a player's performance capability we can use some of the theoretical work of Banister (1991). He theorized performance at any given is determined by the fitness level of the individual less the fatigue.

Therefore by monitoring both fitness and fatigue we can assess when maximal performance is possible and also when maximal performance would be hindered. The measurement of fatigue could be subjective (questionnaire's and scales) or objective (Physiological assessment). It is beyond the scope of this chapter to assess what fatigue is or how it should be assessed. Frequent fitness measurement is also difficult in soccer given the high volume of training and matches. Could training load measures be used to help assess fitness? One of our recent studies (Akubat et al, 2013) showed how integrating the internal and external load could produce ratio's that relate to fitness measures. That is to assess the internal cost of the externally prescribed exercise. This provides a pseudo measurement of exercise economy/efficiency. Regular assessment of external:internal ratio's could give you information on fitness. However unpublished data also shows acute changes in such ratio's with fatigue during competition and between exercise bouts (Akubat et al, unpublished). More reliability and sensitivity studies need to be conducted with such measures but it is an avenue that future research should explore given that both internal and external load is now routinely collected.

To maximize performance we must understand the internal load for the external load prescribed for each individual. Therefore comparing players with each other in this respect may not prove fruitful. A player history or within player comparison would help in assessing this and help us to move from being reactive to proactive. Although relationships we have found of a dose-response nature have been linear more may not always be better. Manzi et al (2009) showed an inverted-U relationship for iTRIMP and HRV changes in runners. They found higher training loads beyond a certain point incurred negative changes in HRV probably a result of overtraining. Therefore through building a player history we can assess what optimum "load" is by comparing periods of different training loads with the performance response. This is an iterative process which would continue to change. The frequency at which this iteration needs to take place depends on the sensitivity of the load measures to physiological change. Maybe the process becomes less iterative where the load measure is more sensitive to changes in athlete physiology, such as in the case of HR where the load adjusts to changes in physiology. An increase in fitness would result in a decreased HR for the same external load meaning a higher external load (intensity or volume) is required to match internal load.

Conclusion

- Most methods of training load measurement have limitations.
- Are you using the one with least limitations within your budget?
- Have you considered the limitations when interpreting the data you have?
- To use training load monitoring effectively understand what the dose means for the individual rather than comparing to players of different physiology.
- We start reactive with the aim of becoming more pro-active based on evidence and a developing player history.
- No golden bullet or ultimate solution yet, but the right information used in the right way could improve decision making for coaches

As Jens Bangsbo says...."Football (Soccer) is not a science, but science can help
Football"

A good knowledge of training load methods, limitations and monitoring methods can make a positive impact on players and their performance and health.

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